TABLE I: ISO/IEC 25010:2023 Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model characteristics, sub-characteristics, definitions, and their proposed applications to EVSE software [1].

Characteristic	Sub-characteristic	Definition†	Application to EVSE Software
Functional Suitability		Degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	Functional suitability may ensure that charging stations perform as expected to meet the needs of EV users. By prioritizing functional suitability, stakeholders can enhance user satisfaction and confidence in the functionality of the charging infrastructure.
	Functional completeness	Degree to which the set of functions covers all the specified tasks and intended users' objectives.	
	Functional correctness	Degree to which a product or system provides accurate results when used by intended users.	
	Functional appropriateness	Degree to which the functions facilitate the accomplishment of specified tasks and objectives.	
Performance Efficiency		Degree to which a product performs its functions within specified time and throughput parameters and is efficient in the use of resources (such as CPU, memory, storage, network devices, energy, materials...) under specified conditions.	Performance efficiency may affect the speed at which vehicles are charged and the overall energy efficiency of the charging process, directly impacting end-users, management system operators, and the load on the power grid.
	Time behavior	Degree to which the response time and throughput rates of a product or system, when performing its functions, meet requirements.	
	Resource utilization	Degree to which the amounts and types of resources used by a product or system, when performing its functions, meet requirements.	
	Capacity	Degree to which the maximum limits of a product or system parameter meet requirements.	
Compatibility		Degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions while sharing the same common environment and resources.	Compatibility may help ensure that charging stations can communicate with various EVs, other charging stations, the power grid, and relevant third-party applications (i.e. management, authentication) via the cloud.
	Co-existence	Degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.	
	Interoperability	Degree to which a system, product or component can exchange information with other products and mutually use the information that has been exchanged.	
Interaction Capability		Degree to which a product or system can be interacted with by specified users to exchange information via the user interface to complete specific tasks in a variety of contexts of use.	Effective interaction capability may aid users in efficiently navigating the charging process, including the human-machine interface (HMI), authentication, and the physical interaction between the charger and the vehicle. This potentially reduces the likelihood of errors and enhances overall user satisfaction.
	Appropriateness recognizability	Degree to which users can recognize whether a product or system is appropriate for their needs.	
	Learnability	Degree to which the functions of a product or system can be learnt to be used by specified users within a specified amount of time.	
	Operability	Degree to which a product or system has attributes that make it easy to operate and control.	
	User error protection	Degree to which a system prevents users against operation errors.	
	User engagement	Degree to which a user interface presents functions and information in an inviting and motivating manner encouraging continued interaction.	

TABLE I (continued).

Characteristic	Sub-characteristic	Definition†	Application to EVSE Software
	Inclusivity	Degree to which a product or system can be used by people of various backgrounds (such as people of various ages, abilities, cultures, ethnicities, languages, genders, economic situations, etc.).	
	User assistance	Degree to which a product can be used by people with the widest range of characteristics and capabilities to achieve specified goals in a specified context of use.	
	Self-descriptiveness	Degree to wich a product presents appropriate information, where needed by the user, to make its capabilities and use immediately obvious to the user without excessive interactions with a product or other resources (such as user documentation, help desks or other users).	
Reliability		Degree to which a system can perform its intended functions under specified conditions for a specified period of time.	Reliability can help to ensure that charging stations function as intended by minimizing downtime and recovering quickly after a disruption of service.
	Faultlessness	Degree to which a system, product or component performs specified functions without fault under normal operation.	
	Availability	Degree to which a system, product or component is operational and accessible when required for use.	
	Fault tolerance	Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.	
	Recoverability	Degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system.	
Security		Degree to which a product or system defends against attack patterns by malicious actors and protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.	Robust security measures are critical to protect user data, the power grid, and the broader EVCS networks connected to the internet [2]. Research focuses heavily on cybersecurity best practices to prevent malicious attacks, which could compromise various components of charging infrastructure [3].
	Confidentiality	Degree to which a product or system ensures that data are accessible only to those authorized to have access.	
	Integrity	Degree to which a system, product or component ensures that the state of its system and data are protected from unauthorized modification or deletion either by malicious action or computer error.	
	Non-repudiation	Degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.	
	Accountability	Degree to which the actions of an entity can be traced uniquely to the entity.	
	Authenticity	Degree to which the identity of a subject or resource can be proved to be the one claimed.	
	Resistance	Degree to which the product or system sustains operations while under attack from a malicious actor.	
Maintainability		Degree of effectiveness and efficiency with which a product or system can be modified to improve it, correct it or adapt it to changes in environment, and in requirements.	

TABLE I (continued).

Characteristic	Sub-characteristic	Definition†	Application to EVSE Software
	Modularity	Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.	Prioritizing maintainability can aid in software being efficiently updated and improved over time, accommodating new technologies and security measures.
	Reusability	Degree to which a product can be used as an asset in more than one system, or in building other assets.	
	Analysability	Degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified.	
	Modifiability	Degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality.	
	Testability	Degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.	
Flexibility		Degree to which a product can be adapted to changes in its requirements, contexts of use or system environment.	Flexibility may allow charging infrastructure to evolve with technological advancements and regulatory changes, especially with the recent push by SAE International to standardize Tesla's North American Charging Standard (NACS) connector [4]. With flexible EVCI, charging station companies can efficiently alter their connectors to be in compliance with NACS SAE J3400 [5].
	Adaptability	Degree to which a product or system can effectively and efficiently be adapted for or transferred to different hardware, software or other operational or usage environments.	
	Scalability	Degree to which a product can handle growing or shrinking workloads or to adapt its capacity to handle variability.	
	Installability	Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.	
	Replaceability	Degree to which a product can replace another specified software product for the same purpose in the same environment.	
Safety		This characteristic represents the degree to which a product under defined conditions to avoid a state in which human life, health, property, or the environment is endangered.	Ensuring safety is paramount to protect users and critical infrastructure from physical hazards. Idaho National Laboratory's "Consequence Driven Cybersecurity for High-Power Charging Infrastructure" details risks associated with physical vulnerabilities in EVCI that lead to "high consequence events," with potential impact to public safety, the power grid, and EV hardware [6]. Reported hazards include cable melting, burn and shock hazards, fires, and brown outs. Prioritizing safety and hazard prevention in EVCI may improve public confidence in the adoption of EVs.
	Operational constraint	Degree to which a product or system constrains its operation to within safe parameters or states when encountering operational hazard.	
	Risk identification	Degree to which a product can identify a course of events or operations that can expose life, property or environment to unacceptable risk.	
	Fail safe	Degree to which a product can automatically place itself in a safe operating mode, or to revert to a safe condition in the event of a failure.	
	Hazard warning	Degree to which a product or system provides warnings of unacceptable risks to operations or internal controls so that they can react in sufficient time to sustain safe operations.	
	Safe integration	Degree to which a product can maintain safety during and after integration with one or more components.	

†Definitions are shown exactly as they are presented in the ISO/IEC 25010:2023 System and Software Product Quality Model [1].

REFERENCES

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